CS 404 – Artificial Intelligence
HW 4 - 2025

100pts

Goal: Guided self-learning and gaining experience with Prolog.

To do:

- **Go to https://swish.swi-prolog.org/**

- Starting with the sample prolog file "gene-hw4.pl" (available under under Lectures/). Load its contents into SWISH (copy-paste on the left side, after choosing new Program).

Add the following predicates with Turkish names in the allocated space (search for TO-COMPLETE). The predicates should be defined such that the following hold among the sample family given below (with the convention of "P(X,Y) holds if X is P of Y"):

Predicates to be added:

- **hala(X,Y)--**aunt on father`s side (sister of Y's father) – e.g. hala(oya,esra) where oya is esra`s aunt.
- **teyze(X,Y)--**aunt on mother`s side (sister of Y's mother)
- **dayi(X,Y)--**uncle on mother`s side (brother of Y's mother)
- **amca(X,Y)--**uncle on father`s side (brother of Y's father)
- **anneanne(X,Y)--**grandma on mother`s side (mother of Y's mother)
- **babaanne(X,Y)--**grandma on father`s side (mother of Y's father)
- **gelin(X,Y)--**daughter-in-law (X is married to Y's son)
- **damat(X,Y)--**son-in-law (X is married to Y's daughter)
- **torun(X,Y)--**grandchild
- **dunur(X,Y) --** (X and Y are dünür if their children are married to each other).

- **list-all-married-couples** —lists all spouses one one line, without repetition (just on eline per couple, see Q2 below)

- **Now add the following sample family** as facts at the end of the file (search for TO-COMPLETE), using the **add_person/5** predicate defined in gene-hw4.pl. You should use the name "single" for spouse name if the person is single and "unknown" if someone's parent name is not known.

  "Ayse has married Ali. They had two children: Osman and Oya. Oya married Murat who has a sister Mualla. Mualla later on married Osman and they had two children, Esra and Elif. Murat's father is Remzi and his wife is Rukiye."

- **Study the codes marked as TO-LEARN or TO-DO. A related question may be asked in the final.**

WHAT TO SUBMIT:

1) **Your updated code as hw4-code-NameLastname.pl**
2) **One pdf to simplify grading, named hw4-answers-NameLastname.pdf**

Include the text of the questions and give your answers afterwards:

- Q1 – 20pts – **What are your newly added Turkish relationship predicates** (there are 10 total)?
- Q2 – 20pts - Write a predicate **list-spouse-pairs** to list all spouses just once and without hitting next (e.g. similar to woman_list predicate we saw in slides). **Include your query and the output of your program** as in Fig. 1(type your query and copy-paste the answers from the output window.
- Q3 - 20pts - Is there someone who has at least two children, who are married (i.e., non-single)? **Include your query and the output of your program** (type your query and copy-paste the answers from the output window.

```
e
?- spouse(X,Y).

X = ali,
Y = ayse
X = ayse,
Y = ali
X = rukiye,
Y = remzi
X = remzi,
Y = rukiye
X = mualla,
Y = osman
X = murat,
Y = oya
```

Fig. 1 Expected query and output answer

- Q4 - 10pts - **Is the order of the clauses of the add-person predicate matter?** (e.g. Can I move add(…) as the last line of the predicate?) Answer as YES/NO and add a one-line explanation.
- Q5 - 30pts - **What is done in mother_check. Explain as:** "it first checks ….. If this succeeds/fails, ….."

HINTS:

- Prolog interprets your code from top to bottom (very important)

- `X @< Y` means: X comes before Y in alphabetical order. *Hint: Can be used in Q2*

- write(X) and write('…') can be used in listing some answers.

- \+ P means "it is not the case that P". *Hint: Used in Q5.*

- You may find it useful **to debug the process** by typing
  trace, (add_person(…)) or trace, (your query here))

in the query window and step-in or over as needed.